Exercise Week 4 Solutions

Advanced Applied Statistics

Fall 2025

Problem 1*

By Bayes' Theorem, $p(\mu|x) \propto p(x|\mu)p(\mu)$, so

$$p(\mu|x) \propto \frac{e^{-\mu}\mu^x e^{-u}}{x!} = \frac{e\mu^{(x+1)-1}e^{-2\mu}}{\Gamma(x+1)}$$

as $\Gamma(x+1) = x!$. Note the resemblance to the pdf of the Gamma Distribution. Indeed,

$$u|x \sim \text{Gamma}(x+1, 2).$$

Problem 2*

(a)

The log-likelihood of π is

$$\ell_x(\pi) = \log[\binom{n}{x}] + x\log\pi + (n-x)\log[1-\pi]$$

$$\Rightarrow \dot{\ell}_x(\pi) = \frac{x}{\pi} - \frac{n-x}{1-\pi} \Rightarrow \ddot{\ell}_x(\pi) = \frac{x}{\pi} - \frac{n-x}{1-\pi}$$

It follows that

$$I(\pi) = -\mathbb{E}[\ddot{\ell}_x(\pi)] = -\mathbb{E}[-\frac{x}{\pi^2} - \frac{n-x}{(1-\pi)^2}] = \frac{\mathbb{E}[x]}{\pi^2} + \frac{n-\mathbb{E}[x]}{(1-\pi)^2} = \frac{n\pi}{\pi^2} + \frac{n-n\pi}{(1-\pi)^2}$$

$$= \frac{n}{\pi} + \frac{n}{1-\pi} = \frac{n}{\pi(1-\pi)}$$

(b)

By definition of Jeffrey's Prior, $p(\pi) = I(\pi)^{\frac{1}{2}}$

$$\Rightarrow p(\pi) = (\frac{n}{\pi(1-\pi)})^{\frac{1}{2}}$$

(c)

Again, we note $p(\pi|x) \propto p(x|\pi)p(\pi)$

$$\Rightarrow p(\pi|x) \propto \binom{n}{x}\pi^x(1-\pi)^{n-x}(\frac{n}{\pi(1-\pi)})^{\frac{1}{2}} \propto \pi^{x-\frac{1}{2}}(1-\pi)^{n-x-\frac{1}{2}}$$

Note the resemblance to the Beta Distribution. Indeed, $\pi|x \sim \text{Beta}(x+\frac{1}{2}, n-x+\frac{1}{2})$.

Problem 3*

(a)

From the provided information,

$$\ell_x(\theta) = p(x = 4|\theta) = \frac{\theta^4 e^{-\theta}}{4!}$$

(b)

Given the prior $p(\theta) = \theta e^{-\theta}$,

$$p(\theta|x) \propto \theta e^{-\theta} \frac{\theta^4 e^{-}\theta}{4!} \propto \theta^6 e^{-2\theta}$$

We again note the resemblance to the Gamma Distribution. Indeed, $\theta|x \sim \text{Gamma}(\alpha = 6, \beta = 2)$.

(c)

Since $\theta|x \sim \text{Gamma}(\alpha = 6, \beta = 2)$, $\mathbb{E}[\theta|x] = \frac{\alpha}{\beta} = \frac{6}{2} = 3$.

(d)

Having seen 4 customers in this hour, our expected average hourly customers increased from 2 to 3.

Problem 4

(a)

$p_A \hat{=} 1,\ \hat{p}_B = 0,\ \hat{p}_C = \frac{1}{2}$.

(b)

Following the hint, $\mathbb{E}[\theta_i] = \frac{1}{2}, \forall i \in \{A, B, C\}$

(c)

First we show $\theta_i|x_i \sim \text{Beta}(\alpha + x_,\beta + 2 - x_i)$.

$$\begin{aligned}
p(\theta_i \mid x_i) &\propto p(x_i \mid \theta_i)\, p(\theta_i) \\
&= \binom{2}{x_i} \theta_i^{x_i}(1-\theta_i)^{2-x_i} \frac{1}{B(\alpha,\beta)} \theta_i^{\alpha-1}(1-\theta_i)^{\beta-1} \\
&\propto \theta_i^{(\alpha+x_i)-1}(1-\theta_i)^{(\beta+2-x_i)-1} \\
&\Rightarrow \theta_i \mid x_i \sim \text{Beta}(\alpha + x_i, \beta + 2 - x_i)
\end{aligned}$$

Recalling the expectation of the Beta Distribution,

$$\mathbb{E}[\theta_i|x_i] = \frac{\alpha + x_i}{2 + \alpha + \beta}$$

(d)

Recall that $\hat{p}_i = \frac{x_i}{2}, \mathbb{E}[\theta_i] = \frac{\alpha}{\alpha+\beta}$ We can rewrite the empirical Bayes posterior mean as

$$\mathbb{E}[\theta_i|x_i] = \frac{x_i}{2+\alpha+\beta} + \frac{\alpha}{2+\alpha+\beta} = (\frac{2}{2+\alpha+\beta})\hat{p}_i + (\frac{\alpha+\beta}{2+\alpha+\beta})\mu$$

where $\mu = \mathbb{E}[\theta_i]$. In this form, we see that the posterior mean is the weighted average of the prior mean and the estimated probability of solving correctly. The higher $\alpha + \beta$, the more a student's average is pulled towards the prior mean.